

VIRAJ VAGHASIA

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M.S. Computer Science candidate (May 2027) with three years of production software engineering experience, building backend services, data pipelines, and document retrieval systems.

EDUCATION

University of Southern California , Los Angeles, CA	Aug 2025 - May 2027
Master of Science in Computer Science	GPA: 3.88/4.00
Dwarkadas J. Sanghvi College of Engineering , Mumbai, India	Aug 2018 - May 2022
Bachelor of Engineering, Information Technology	GPA: 9.38/10
Coursework: Analysis of Algorithms, Information Retrieval, Database Systems, Applied Natural Language Processing	

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, Bash

Backend & Data: FastAPI, Django, Flask, Node.js, Express, REST APIs, PostgreSQL, MySQL, MongoDB, Redis, SQLite, ETL and batch pipelines

Frontend: React, Next.js, Vue.js, Redux, Tailwind CSS

Infrastructure & Testing: Docker, AWS, GitHub Actions (CI/CD), Git, pytest, Playwright, Modal

Search & Retrieval: Dense embeddings, BM25, hybrid ranking, cross-encoder reranking, evaluation harnesses, Qdrant, vLLM, Ollama

EXPERIENCE

Research Assistant, USC Information Sciences Institute	May 2026 - Present
<i>DARPA CRITICALMAAS - geological literature acquisition and classification pipeline, Los Angeles, CA</i>	
- Built a restartable document-acquisition service in Python checkpointed on a SQLite ledger rather than a run flag, pairing an open-access tier with an authenticated Playwright tier through the USC library proxy; the ledger records 30,096 downloaded articles across 9 journals. - Replaced per-publisher scrapers with declarative JSON recipes covering 4 platforms, plus a generator that proposes extraction rules from one open-access article - deterministic checks decide what ships, not the model. - Classified crawl failures into login redirects, bot challenges and rate-limit blocks instead of retrying blindly, with escalating 3-to-30-minute backoff and a halt after three consecutive blocks - an institutional IP block is terminal, not retryable. - Diagnosed silent data-loss defects - discarded supplementary files, an ignored year filter - by comparing files on content hash after filename comparison produced false mismatches, and grew the pytest suite to 295 tests.	
Senior Software Engineer, Decimal Point Analytics, Mumbai, India	Jul 2022 - Jul 2025
- Developed a real-time financial index platform in React and FastAPI tracking multiple indices simultaneously, replacing static analyst reports with interactive time-series views. - Implemented a maker-checker review workflow in Django separating the analyst entering data from the reviewer approving it, blocking unreviewed records from publication. - Led a Loan Management System and an ESG questionnaire portal across several engineers, onboarding new joiners and gathering requirements from analysts; designed the questionnaire schema so business users could change questions without a code deploy (Vue.js, Next.js, FastAPI).	

PROJECTS

SQLAlchemy Upgrade Agent Python, FastAPI, Qdrant, GitHub Actions, Modal	Jun 2026 - Sep 2026
<i>virajvaghasia--sqlalchemy-upgrade-agent.modal.run github.com/virajvaghasia/sqlalchemy-upgrade-agent</i>	
- Shipped a deployed question-answering service over SQLAlchemy 1.4 to 2.0 migration documentation that cites a source page for every claim and declines when the retrieved pages do not answer. - Built hybrid retrieval over 3,284 passages - dense embeddings with BM25, duplicate collapsing, cross-encoder reranking - raising correct-page-in-top-5 from 0.49 to 0.64 on a 100-question hand-verified set, measured as paired wins rather than an average. - Verified answers by executing each claim against SQLAlchemy 2.0.51 instead of trusting a model's self-report (47 of 51 correct), and gated retrieval quality in GitHub Actions - it blocked a pull request removing the reranker. - Rejected two proposed retrieval improvements on their own measurements - one dropped recall from 0.64 to 0.58 and was reverted, the other cancelled before implementation once the defect it assumed was shown not to exist.	